

Follow the Leader: Agenda-Setting and Strategic Complementarity in Corporate Lobbying Networks

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Abstract. [Placeholder: 70–150 word summary of the paper.]

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1 Background and Literature Review

2 Methodology

2.1 Data Source

The Lobbying Disclosure Act (LDA) [16] requires all organizations who engage in lobbying for federal legislation to publicly disclose these activities every fiscal quarter. We used bulk lobbying data from OpenSecrets [6] to access LDA reports. We focused on the 111th through 117th Congresses, spanning from January 2009 through January 2023. We chose this period because the data were available through OpenSecrets’ free bulk disclosure files and because the quarterly reporting standards introduced by the Honest Leadership and Open Government Act of 2007 [12] remained consistent throughout, enabling methodological comparability across sessions. Our specific session of focus, however, was the 116th Congress, which ran from January 2019 through January 2021. We selected the 116th Congress as our primary analytical window because it spans both a stable pre-pandemic period and the onset of COVID-19, producing substantive variation in legislative priorities useful for testing the scope of our influence measure.

2.2 Data Processing

Using the OpenSecrets bulk lobbying data, we extracted valid LDA reports following the criteria in the OpenSecrets Data User Guide [7]. We retained only reports flagged with the validity indicator *ind* = y, which identifies the final, correct filing for each report and removes superseded amendments. We additionally retained only reports listing at least one named lobbyist.

We filtered the sample to Fortune 500 companies using the 2019 Fortune 500 ranking (from the beginning of the 116th Congress). Kowal [13] similarly restricts analysis to Fortune 500 firms, arguing that they constitute a representative cross-industry sample of the largest U.S. corporations with the most consistently disclosed lobbying activity, and that data for smaller firms are not publicly available in sufficient detail. Matching LDA records to Fortune 500 firms requires resolving the many client name variants, subsidiary names, and registrant names under which these large corporations file. We accomplished this through a custom mapping applied to the `ultorg` and `client` fields in the OpenSecrets data, enabling us to identify all valid LDA activity by Fortune 500 companies and subsidiaries across congressional sessions.

The result was a dataset with company, bill, and expenditure amounts as the primary dimensions. For each LDA report, we disaggregated the reported dollar amount evenly across all bills named in that report.

2.3 Bill Affiliation Network

To understand the co-lobbying structure among Fortune 500 firms and to establish the industry communities that anchor our within-community analyses, we constructed a bill affiliation network. This is an undirected weighted graph in which each node is a Fortune 500 firm and each edge weight counts the number of distinct bills co-lobbied by a pair of firms. The affiliation network does not encode direction or prioritization. It captures the degree to which two firms share a common legislative agenda broadly speaking.

We applied the Leiden community detection algorithm [20] to this network to partition firms into lobbying communities. Using the default resolution parameter $\gamma = 1.0$, which maximized interpretable industry alignment, we obtained six lobbying communities that closely correspond to established industry groupings. These community labels are used throughout to contextualize firm-level results.

To benchmark our novel agenda-setting measure against prior network-based approaches to corporate influence, we also computed three centrality measures from the affiliation network: global PageRank, Katz-Bonacich centrality, and within-community eigenvector centrality for each set of firms in the same Leiden lobbying communities. Mizruchi and Bunting [18] demonstrate that centrality measures of this type predict influence outcomes in corporate networks better than raw interlock counts. We compared the relationship between these measures of network structure to our directed agenda-setting measure later in our analysis.

2.4 Influence Network Creation

Our primary focus was to construct a directed influence network among Fortune 500 lobbying firms, focusing specifically on lobbying activity in the 116th Congress.

We sought to use agenda-setting ability as a proxy for influence. Scott [19] demonstrates that corporate lobbying influence operates through agenda overlap, distinguishing true peer influence from bandwagon behavior. Scott finds that

firms that are influential systematically shape which bills their peers prioritize, not merely which bills become popular. Hall and Deardorff [10] formalize lobbying as a legislative subsidy, in which firms supply policy-relevant information to legislators to advance shared legislative priorities. Under this framework, the firms that act first on a bill hold an informational advantage that translates into agenda-setting influence over peers with aligned interests. This operationalization is distinct from centrality-based measures of influence, which capture structural position in an undirected co-lobbying network but cannot identify which of two structurally similar firms leads and which follows (Mizruchi and Bunting, 1981; Carpenter, Esterling, and Lazer, 2004).

To operationalize this, we constructed an RBO-weighted directed network with directed edges based on temporal first-mover precedence across the eight quarters of a congressional session. The total edge weight between each pair of companies represents their Rank-Biased Overlap (RBO), a measure of the similarity in their ranked lobbying bill portfolios based on expenditure [23]. Each pair of firms is connected by two directed edges (one in each direction), whose weights distribute the pair’s RBO score proportionally by first-mover advantage. The asymmetry in edge weights between the two directions of a pair reflect first-mover advantage: a stronger directed edge $i \rightarrow j$ indicates that firm i lobbied a greater share of the pair’s shared bills before firm j across the congressional session. RBO is an effective measure because it captures both the overlap in the set of bills a pair lobby for and how similarly they budget across those bills. For example, two firms may co-lobby the same set of bills while concentrating spend on entirely different priorities. RBO makes this comparison possible by operating on each firm’s ranked bill list, weighting agreement at the top of the ranking more heavily than agreement further down [23]. The RBO edge weight between two companies thus reflects the similarity in their lobbying priorities, not merely their lobbying activity.

The construction proceeded as follows. For each Fortune 500 firm, we computed the fraction of total 116th Congress lobbying spend allocated to each bill and ranked bills in descending order of that fraction, retaining the top 30 bills per firm. This produced a ranked priority list L_i for each firm i . For every pair of firms i, j sharing at least one bill, we computed the Rank-Biased Overlap [23] of their ranked lists as follows:

$$\text{RBO}(L_i, L_j; p) = (1 - p) \sum_{d=1}^k p^{d-1} \cdot \frac{|L_i^{(d)} \cap L_j^{(d)}|}{d} \quad (1)$$

where $L^{(d)}$ denotes the top- d elements of a ranked list and p is a persistence parameter controlling the weight placed on top-ranked elements. We set $p = 0.85$, calibrated against the empirical spend concentration of Fortune 500 firms. At this value, RBO assigns approximately 65% of its weight to the top seven bills, matching the interquartile range (P25-P75) of cumulative budget allocation across bill ranks in the data.

We then determined temporal precedence for each shared top-30 bill. For each shared bill b , we identified the first quarter in which each firm’s lobbying

activity on b appears. If firm i first appeared on bill b in an earlier quarter than firm j , firm i received one first-mover credit for that bill. Ties (bills where both firms first appeared in the same quarter) were split equally, each receiving one-half of a first-mover credit. Aggregating over all shared top-30 bills, the net lead count is:

$$\begin{aligned} \text{net_temporal}(i, j) = \sum_{b \in L_i \cap L_j} & \mathbf{1}[\text{quarter}_i(b) < \text{quarter}_j(b)] \\ & - \mathbf{1}[\text{quarter}_j(b) < \text{quarter}_i(b)] \end{aligned} \quad (2)$$

For every firm pair with $\text{RBO} > 0$, both directed edges are emitted. Let $F_i(j)$ denote the number of shared bills on which i lobbied first, $F_j(i)$ the number on which j lobbied first, $T(i, j)$ the number of ties, and $|L_i \cap L_j| = F_i(j) + F_j(i) + T(i, j)$ the total shared bill count. The directed edge weight from i to j is:

$$w(i \rightarrow j) = \frac{F_i(j) + \frac{1}{2} T(i, j)}{|L_i \cap L_j|} \cdot \text{RBO}(L_i, L_j; p) \quad (3)$$

By construction, $w(i \rightarrow j) + w(j \rightarrow i) = \text{RBO}(L_i, L_j; p)$, so each pair's RBO score is fully partitioned between the two directed edges proportionally to first-mover credits. Ties cancel in the net calculation: $\text{net_temporal}(i, j) = F_i(j) - F_j(i)$ defined above captures the asymmetry that drives net_strength .

The primary node-level measure is net agenda-setting strength:

$$\begin{aligned} \text{net_strength}(i) &= \sum_j [w(i \rightarrow j) - w(j \rightarrow i)] \\ &= \sum_j \frac{F_i(j) - F_j(i)}{|L_i \cap L_j|} \cdot \text{RBO}(L_i, L_j; p) \end{aligned} \quad (4)$$

Ties cancel in the net calculation, so net_strength depends only on the difference in directed lead counts across all firm pairs. A positive net_strength identifies a net agenda-setter; a negative value identifies a net follower. We use net_strength as the primary measure of influence throughout this paper. For within-community analysis, we compute $\text{wc_net_strength}(i)$ by restricting the sum to firm pairs within the same Leiden community.

2.5 Interpreting the Influence Network

Under this model of influence, the weight of a directed edge reflects the first-mover advantage of one company lobbying before another on the same bills. An edge pair with a high RBO means that the two firms not only lobby for similar bills but also prioritize similar bills in their spending. Scott [19] makes precisely this distinction, arguing that it is the alignment of priorities, not merely shared

bill participation, that constitutes peer influence and separates agenda-setters from bandwagon followers.

If one firm in the relationship is the dominant first-mover, they exert greater influence. Their outgoing edge weight equals the proportion of shared bills on which they held first-mover advantage (with ties split equally), multiplied by the pair’s RBO score. Another way to conceptualize this directed network, which more directly speaks to the first-mover advantage, is that a net follower amplifies the importance of the bills the agenda-setter initially lobbies for. This gives first movers a powerful opportunity to shape which bills receive coalition attention and to encourage similar firms to join them, consistent with the legislative-subsidy model of Hall and Deardorff [10].

2.6 Filtering

One of the key design decisions we made was to exclude bills lobbied by more than 50 Fortune 500 companies. We chose this threshold to avoid inflating our similarity metric. For example, the CARES Act, an omnibus COVID-relief bill, was lobbied for by 170 of the Fortune 500 companies in 2019, which overshadows the goal of understanding strategic similarities and influence between firms. Lobbying on these high-participation bills reflects what Scott [19] describes as bandwagon behavior rather than genuine agenda-setting influence. The threshold of 50 was selected at a natural break in the bill-prevalence distribution.

2.7 Summary Statistics

In total, for the 116th Congress, 381 Fortune 500 companies (and their subsidiaries) lobbied for 2,245 bills and collectively spent \$1.73 billion in lobbying expenditure across the eight quarters. After applying the mega-bill filter, the RBO directed influence network contains 293 firms and 3,869 RBO-linked company pairs, each represented by two directed edges whose weights sum to the pair’s RBO score. Among these pairs, 48.8% (1,890) have asymmetric edge weights in which one firm holds a clear first-mover advantage across their shared agenda (*net-temporal* $\neq 0$); the remaining 1,979 pairs have all shared bills tied, yielding equal edge weights in both directions. The cross-congressional stability analyses use equivalent per-congress networks over the 111th through 117th Congresses, with a stable subset of 166 firms lobbying across all seven sessions.

2.8 Analyses

Using the directed influence network, we conducted a series of analyses to reveal patterns in Fortune 500 lobbying behavior and to assess evidence of strategic complementarity. We first characterized the community structure of the bill affiliation network to establish the industry communities that contextualize all firm-level results. We then identified the top global and within-community agenda-setters and followers in the 116th Congress, examining the relationship between

agenda-setting strength, lobbying expenditure, and portfolio breadth. Next, we assessed the cross-congressional stability of the influence hierarchy across the 111th through 117th Congresses, testing both within-community leadership persistence and global top-set rotation. We then tested for strategic complementarity through a payoff-response regression, a direction consistency analysis, and a bill adoption diffusion analysis. Finally, we compared our agenda-setting measure to structural centrality measures from the affiliation network to demonstrate what the directed influence network captures that structural position alone does not.

3 Results

3.1 Community Structure

The bill affiliation network for the 116th Congress comprises 281 firms connected by co-lobbying activity across 2,245 bills. Applying the Leiden community detection algorithm [20] at resolution parameter $\gamma = 1.0$, we identify six communities: Finance/Insurance (72 firms), Tech/Telecom (55), Energy/Utilities (49), Defense/Industrial (46), Healthcare/Pharmaceuticals (42), and Consumer/Manufacturing (17). These communities closely reflect established industry clusters, and they serve as the reference partition for within-community analyses throughout this paper.

The community structure is broadly consistent with prior work on co-lobbying affiliation networks, which consistently finds that firms cluster by industry sector [5, 15]. Cross-community ties are nevertheless present: Berkshire Hathaway, classified in the Energy/Utilities community as a large energy holding company, holds a directed edge to CSX (Defense/Industrial) as its strongest single follower, reflecting shared lobbying on infrastructure and rail-adjacent legislation. Firms that bridge communities in this way tend to score higher on global *net_strength*, consistent with arguments that cross-sector positions confer structural advantages in lobbying coalitions [11].

3.2 Primary Directed Influence Network: Top Agenda-Setters

Using the directed agenda-setting network for the 116th Congress, we identify the top 30 firms by global *net_strength*, our primary measure of lobbying influence. The network contains 293 firms (of which 281 are connected) and 7,738 directed edges derived from 25,676 valid LDA reports.

Energy/Utilities firms dominate the global ranking. DTE Energy ranks first with *net_strength* = 5.843, followed by Calpine (4.943), AbbVie (4.167), CMS Energy (3.889), and Berkshire Hathaway (3.776). The dominance of utility companies at the top of the global ranking reflects the dense, overlapping structure of regulatory legislation in that sector. A small cluster of utilities consistently lobbies for the same infrastructure, grid upgrades, and environmental compliance bills, and a few firms systematically file before their peers. DTE Energy ranks

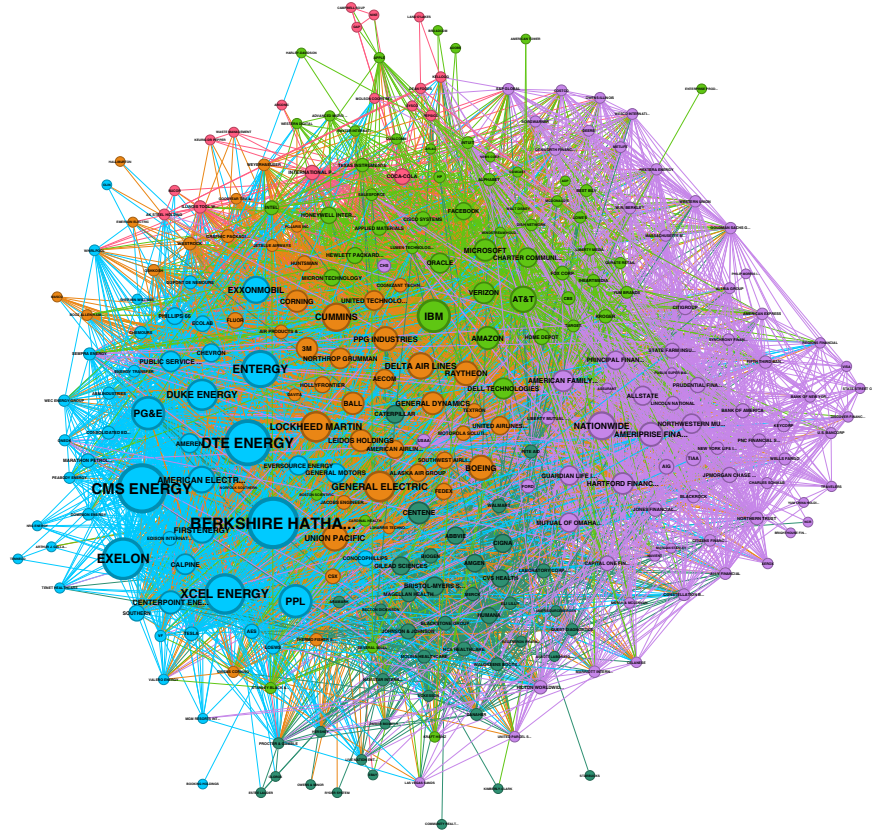


Fig. 1. Bill affiliation network for the 116th Congress. Nodes represent Fortune 500 firms, sized by total lobbying expenditure and colored by Leiden community assignment. Edges encode co-lobbying on shared bills.

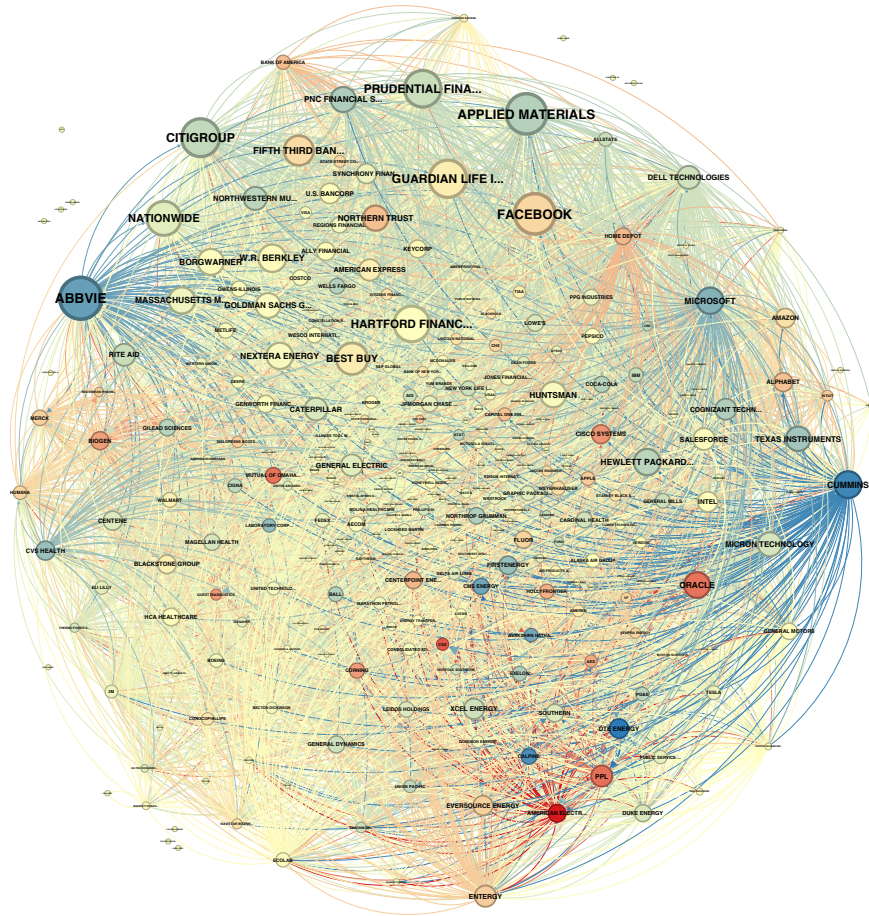


Fig. 2. Full directed RBO influence network for the 116th Congress (293 firms, 7,738 directed edges). Node color encodes global *net_strength* rank (darker = higher rank); node size encodes weighted out-degree.

285th of 293 firms by total lobbying spend while ranking first by *net_strength*, illustrating a central implication of our measure: early, targeted lobbying on bills that many similar firms subsequently adopt demonstrates agenda-setting influence that cannot be recovered from expenditure data alone.

In this way, *net_strength* is not well predicted by total lobbying expenditure. We find a weak negative Spearman correlation between *net_strength* and total spend ($\rho = -0.136$, $p = 0.020$) and a weak positive correlation with portfolio breadth measured as number of bills lobbied ($\rho = +0.188$, $p = 0.001$). The list of top-30 firms by *net_strength* overlaps with the top-30 by total spend in only 2 of 30 firms, and with the top-30 by number of bills in only 8 of 30. De Figueiredo and Richter [9] note the difficulty of using lobbying expenditure as a proxy for effectiveness, since a disperse high-spend lobbying portfolio may produce less influence per dollar than a concentrated and timely one. Our results quantify this gap directly: among the 30 largest agenda-setters in the 116th Congress, most are not among the highest spenders, and several, such as DTE Energy, CMS Energy, and Texas Instruments, rank in the bottom half of the Fortune 500 list by total expenditure.

Within each of the six Leiden communities, we identify within-community leaders by *wc_net_strength*. The 116th Congress community leaders are: PNC Financial Services, Wells Fargo, Citigroup (Finance/Insurance); Microsoft, Texas Instruments, Hewlett Packard Enterprise (Tech/Telecom); DTE Energy, Calpine, CMS Energy (Energy Utilities), Northrop Grumman, Ball, Norfolk Southern (Defense/Industrial), AbbVie, Laboratory Corp. of America, CVS Health (Healthcare/Pharmaceuticals); and AK Steel Holding, Coca-Cola, Gap (Consumer/Manufacturing). Several firms appear in both the global and within-community top tiers. Microsoft, AbbVie, Northrop Grumman, and DTE Energy are simultaneously the top agenda-setters within their respective communities and among the top 30 globally. Others, such as Cummins (global rank 2, within-community rank 10 in Defense/Industrial), achieve high global *net_strength* primarily through cross-community influence rather than community-level leadership.

The top-20 influencer-follower pairs by directed edge weight are predominantly within-community, with a few notable cross-community exceptions. The highest-weight pair is Berkshire Hathaway $\rightarrow$ CSX (RBO weight 3.266, 17 shared bills), a cross-community pair in which an energy holding company set the lobbying priorities for a freight railroad on infrastructure and energy-adjacent legislation. Within Healthcare/Pharmaceuticals, Laboratory Corp. of America leads Quest Diagnostics (1.786) and AbbVie leads Merck (1.798). Within Tech/Telecom, Microsoft leads Oracle (1.786) and CBS leads Fox Corp. (1.591). These pairs illustrate the industry-specific logic of agenda-setting: firms operating in the same regulatory environment tend to share the strongest directed relationships, and the firms that consistently file first are those that supply policy-relevant information early enough to shape their peers' legislative priorities, consistent with the legislative subsidy framework of Hall and Deardorff [10].

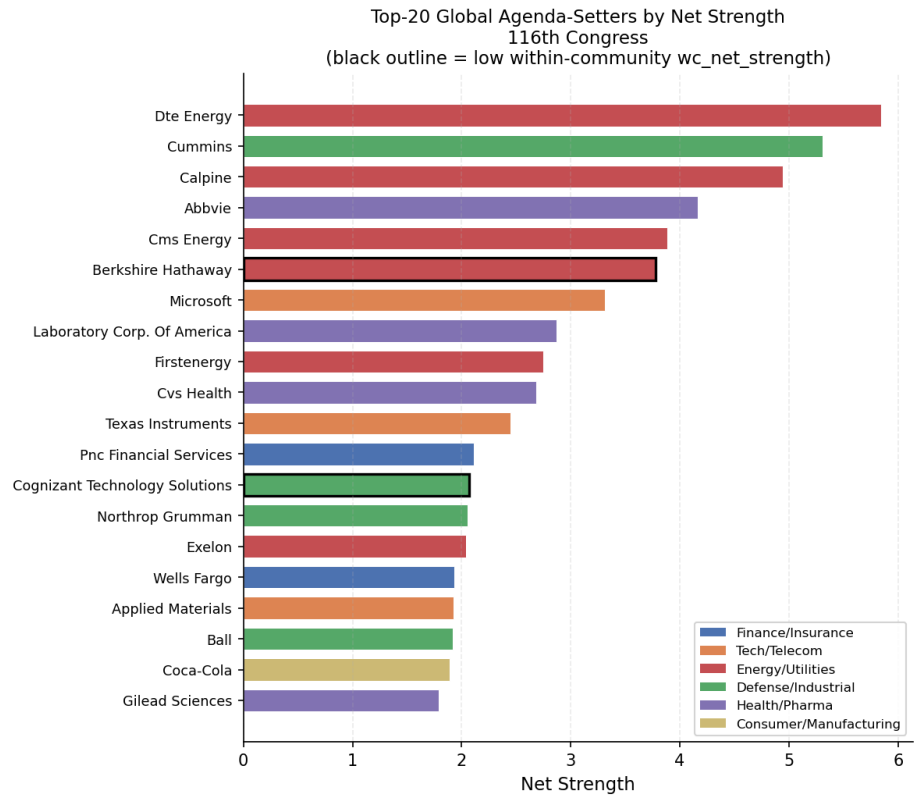


Fig. 3. Top-20 global agenda-setters by *net_strength* in the 116th Congress. Firms simultaneously ranking among the lowest within-community *wc_net_strength* values are highlighted.

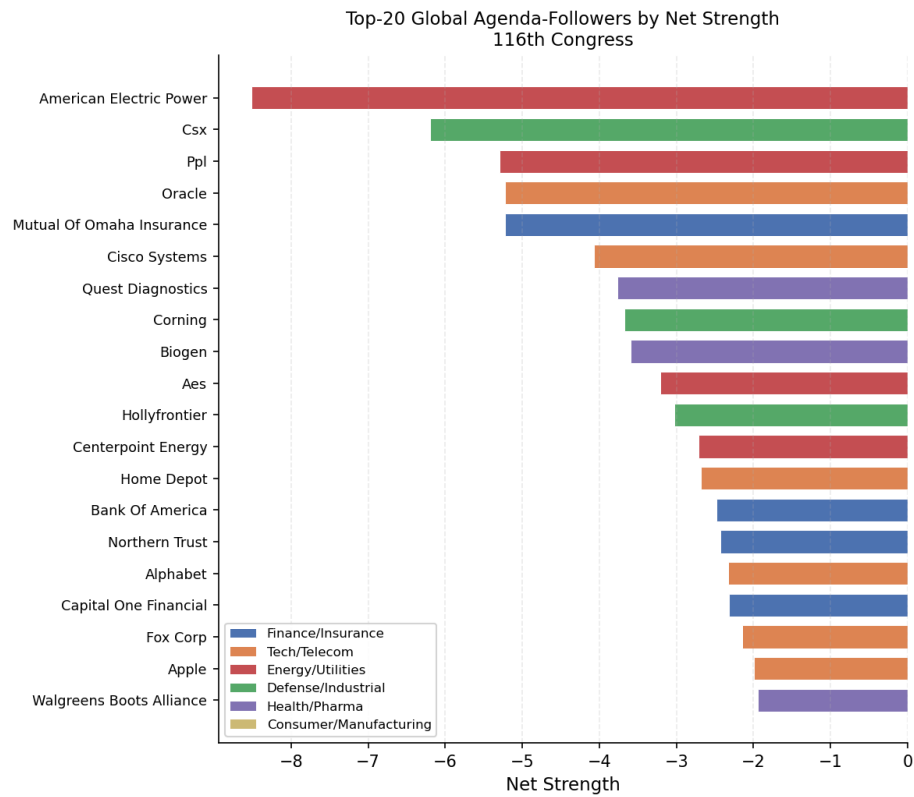


Fig. 4. Top-20 global agenda-followers (lowest *net_strength*) in the 116th Congress.

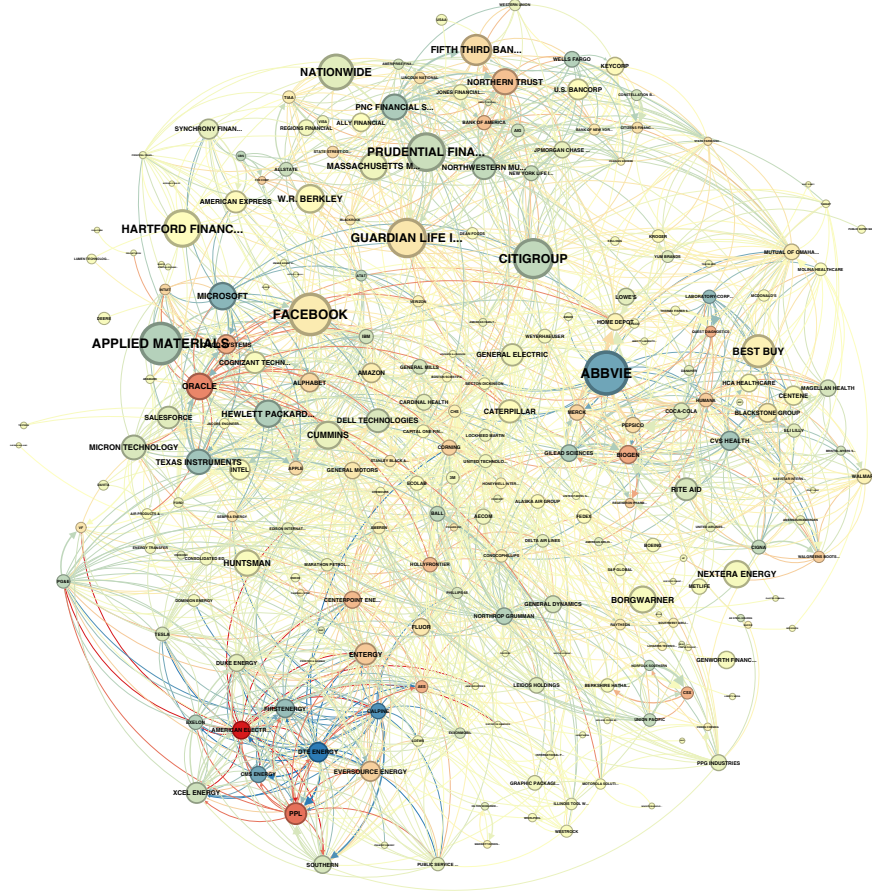


Fig. 5. Filtered directed influence network retaining the highest-weight edges. Node color encodes global *net_strength*; node size encodes weighted degree.

3.3 Cross-Congressional Industry Hierarchy

To examine whether within-community leadership structures persist across congressional sessions, we compute *wc_net_strength* for firms present in all seven congresses (111th through 117th) and assess rank concordance using Kendall’s coefficient of concordance W . A significant positive W indicates that the same firms tend to occupy the top within-community positions session after session.

The results differ markedly across communities. Energy/Utilities exhibits the strongest concordance across congressional sessions ($W = 0.446$, $p < 0.001$), followed by Tech/Telecom ($W = 0.334$, $p < 0.001$), Defense/Industrial ($W = 0.299$, $p = 0.002$), and Finance/Insurance ($W = 0.242$, $p = 0.007$). Healthcare/Pharmaceuticals is the only community with a statistically non-significant concordance ($W = 0.088$, $p = 0.858$). Consumer/Manufacturing contains too few stable firms for a reliable rank concordance estimate.

Using a threshold of top-5 within-community rank in at least 4 of 7 congresses, we identify the following persistent community leaders. In Tech/Telecom: Microsoft (5 of 7 congresses: 111th, 112th, 115th, 116th, 117th) and Texas Instruments (5 of 7: 113th–117th). In Energy/Utilities: Xcel Energy (6 of 7), Duke Energy (5 of 7), DTE Energy (5 of 7), and Exelon (4 of 7). In Defense/Industrial: Lockheed Martin (5 of 7) and United Airlines Holdings (5 of 7). No firm in Finance/Insurance or Healthcare/Pharmaceuticals reaches this threshold, meaning that within-community leadership in those sectors turns over substantially from session to session.

The sectoral pattern is consistent with prior accounts of lobbying coalition dynamics. The absence of persistent leaders in Healthcare/Pharmaceuticals aligns with Heaney’s [11] analysis of that sector, which highlights frequent reshuffling of coalitions around shifting legislative priorities and the disproportionate role of trade associations. Finance/Insurance shows a similarly fragmented pattern. The top within-community agenda-setters differ substantially between the 111th Congress following the banking crisis compared to the deregulatory sessions that followed. LaPira et al.’s [15] distinction between institutionally embedded, persistent lobbying actors, and issue-driven actors explains this sectoral divide: the firms that appear as persistent community leaders in Energy and Defense – sectors governed by long-horizon regulatory cycles – resemble the persistent actors in their framework, while the rotating set of Healthcare/Pharmaceuticals and Finance/Insurance leaders corresponds to the transient pattern they document.

3.4 Global Top-30 Leader Rotation

At the global level, cross-congressional stability is more limited. Among the 166 firms present in all seven sessions, the mean Spearman rank correlation between consecutive *net_strength* rankings is only $\rho = 0.145$, reaching statistical significance in 4 of 6 consecutive session pairs. The Jaccard similarity of top-30 sets averages 0.167, ranging from 0.132 to 0.250. On average, roughly 7 to 12 of the top 30 agenda-setters in any given session also appear in the top 30 of the following session. No firm appears in the top-30 agenda-setters across all seven

sessions. Among the 166-firm stable cohort across congresses, the firms with the highest mean *net_strength* across all seven sessions are Xcel Energy (mean = 4.892), Lockheed Martin (4.565), Duke Energy (4.449), Microsoft (3.604), Northwestern Mutual (3.333), and DTE Energy (3.331).

The substantial top-30 turnover from session to session suggests that the identity of top global agenda-setters is partially issue-driven. Each congressional cycle introduces a distinct set of legislative priorities, and firms positioned around those specific bills emerge as top influencers for that session before ceding their position in the next. The marked drop in consecutive-session similarity between the 115th and 116th Congresses (RBO = 0.053; Jaccard of top-30 = 0.132) coincides with the onset of COVID-19, which reshuffled lobbying priorities broadly and produced a largely new cohort of top global agenda-setters in the 116th Congress, including AbbVie, Calpine, and Ball, none of whom appeared in the previous session’s top 30.

3.5 Strategic Complementarity

To assess whether the structure of the directed influence network is consistent with the strategic payoff complementarity proposed by Ballester et al. [1], we estimate the following interaction specification:

$$\Delta \log \text{spend}_{i,b,t+1} = \beta_1 \text{entry}_j + \beta_2 \text{rbo}_{ij} + \beta_3 (\text{entry}_j \times \text{rbo}_{ij}) + \gamma_{ib} + \delta_t + \varepsilon_{ibt} \quad (5)$$

Here, *entry_j* indicates that firm *j* first lobbied for bill *b* in the current quarter, and *rbo_{ij}* is the ranked portfolio similarity between firms *i* and *j*. The terms γ_{ib} , δ_t , and ε_{ibt} are firm-bill fixed effects, quarter fixed effects, and the idiosyncratic error, respectively. Firm-bill fixed effects γ_{ib} absorb all time-invariant heterogeneity in firm *i*’s engagement with bill *b*, including baseline propensity to lobby on that bill and the bill’s inherent salience to the firm. Quarter fixed effects δ_t control for aggregate trends in lobbying spending common to all firm-bill pairs in a given reporting period. A positive and significant β_3 implies that firm *i* increases its spend on bill *b* more when a more portfolio-similar peer first enters that bill, consistent with strategic complementarity. The full-panel estimate yields $\beta_3 = -0.125$ ($p < 0.001$), indicating that on average the spending response to a co-lobbyist’s entry is muted for similar pairs. This full-panel result, however, conceals substantial heterogeneity across the RBO distribution.

Restricting the sample to pairs with RBO at or above the 75th percentile, we find $\beta_3 = +0.147$ ($p = 0.003$). These results show evidence of strategic complementarity: firms with closely aligned portfolios and priorities increase spending in response to a peer’s bill entry. Restricting to pairs below the 25th percentile (with RBOs below 0.007), we find $\beta_3 = -8.00$ ($p < 0.001$), consistent with strategic substitution. Together, these results indicate that strategic complementarity is conditional on priority alignment. Fortune 500 firms with similar lobbying priorities amplify each other’s activity when a shared bill is entered, while firms with divergent portfolios reduce theirs.

Ward [22] shows that in a formal lobbying game, strategic offsetting, which is analogous to the substitutive behavior observed in low-RBO pairs, is a general equilibrium feature that arises from the geometry of vector position-taking, independently of any complementarity parameter. Our analysis based on Ballester et al. [1] refines this prediction: the direction of the strategic response depends on lobbying agenda similarity, with complementarity emerging only among firms already aligned in their legislative priorities.

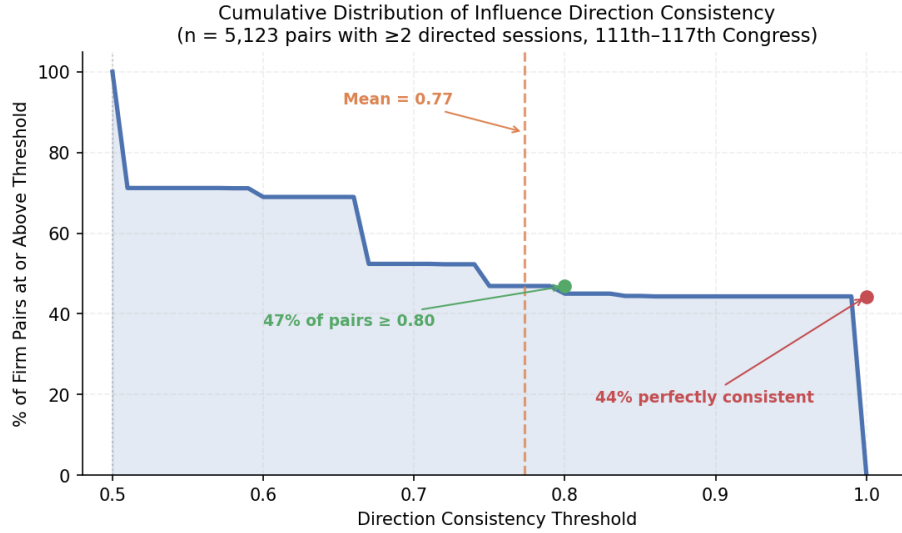


Fig. 6. Edge direction consistency across the 111th–117th Congresses. Among 5,123 firm pairs appearing as decisive in at least two sessions, the mean direction consistency is 0.773 (median 0.750); 44.3% maintain perfectly consistent direction in every co-appearing session.

We also examine direction consistency across congressional sessions. Among the 5,123 firm pairs that appear as decisive in at least two sessions, the mean direction consistency (the fraction of sessions in which the same firm holds the influencer role) is 0.773, with a median of 0.750. Of these pairs, 44.3% maintain perfectly consistent direction across every session in which they co-appear, and 71.2% show majority-direction consistency. The persistence of directed relationships reinforces the view that the agenda-setting directions we recover reflect durable structural properties of firm-level lobbying behavior. Firms that set the agenda within their communities do so repeatedly, and the follower-influencer relationship between a given pair is largely stable over the seven congressional sessions we examine. Box-Steffensmeier et al. [2] find similarly persistent coalition structures in legislative cue-taking, suggesting that the stability we observe at the firm level is part of a broader pattern of durable relationships in the

lobbying ecosystem. Even though the top set of agenda-setters may evolve between congressional sessions, the direction of influence between aligned pairs stays stable.

3.6 Centrality Measures and Agenda-Setting

Prior work on corporate networks has used structural centrality as a proxy for firm influence, such as Mizruchi and Bunting [18] demonstrating that centrality in corporate interlock networks predicts political influence better than raw interlock counts. Furthermore, Chu and Davis [8] document the declining predictive power of board interlocks over time, motivating a search for alternative influence mechanisms. To assess whether our directed *net_strength* measure is essentially the same as structural network centrality, we compute three centrality measures from the bill affiliation network: global PageRank, Katz-Bonacich centrality, and within-community eigenvector centrality, and examine their Spearman correlations with *net_strength* and *wc_net_strength*.

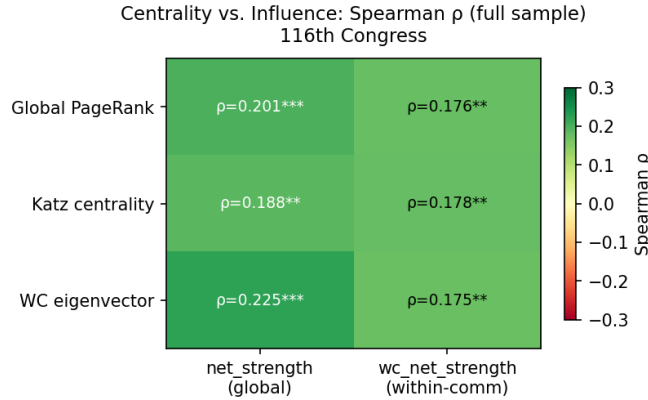


Fig. 7. Spearman correlation heatmap between *net_strength*, *wc_net_strength*, global PageRank, Katz-Bonacich centrality, and within-community eigenvector centrality for the 116th Congress.

The correlations are positive and statistically significant but weak. Within-community eigenvector centrality exhibits the strongest association with global *net_strength* ($\rho = 0.225$, $p < 0.001$) and with *wc_net_strength* ($\rho = 0.175$, $p = 0.006$). Global PageRank reaches $\rho = 0.201$ ($p < 0.001$) against global *net_strength*. The top-30 firm overlap between any centrality ranking and the *net_strength* ranking reaches at most 37%, for global PageRank.

These results show that a firm can be highly central in the bill affiliation network, co-lobbying on many bills that others also pursue, without emerging as a temporal agenda-setter. Structural position and temporal precedence are

distinct dimensions of lobbying influence that co-occurrence networks cannot disentangle. A centrality-based analysis of the same 116th Congress data would misclassify over 60% of the top agenda-setters identified by *net_strength* and would surface a different set of firms whose visibility reflects breadth of legislative activity rather than early, systematic prioritization that spreads across the network. Thus, the directed RBO network, constructed from quarterly first-mover information across eight periods within the session, recovers a dimension of lobbying behavior that static undirected measures cannot observe.

3.7 Shared Lobbyist Mechanism

One mechanism proposed for the transmission of legislative priorities across firms is direct intermediary sharing: two firms that employ the same individual lobbyist or the same lobbying firm may converge on a shared bill agenda as a result of that shared contractor [5]. We test this by asking whether influencer-follower pairs are more likely to share a lobbyist or lobbying firm on the specific bills where one leads the other.

The evidence against bill-level intermediary mediation is clear. Among 3,347 decisive pairs (those where the influencer demonstrably files before the follower on a shared bill) only 7 share a specific named lobbyist or lobbying firm on that bill. The shared-lobbyist mediation rate is below 0.5% across all RBO quartiles, indicating that the pattern does not differ between high- and low-similarity pairs. In fact, instances with detectable shared affiliation are attributable to structural anomalies rather than genuine bilateral coordination. The shared contractor between United Technologies and Raytheon, for example, is explained by their April 2020 merger, which made them a single corporate entity within the congressional session.

These findings suggest that the agenda-setting patterns we identify are not primarily driven by observable intermediary relationships. Rather, agenda-setting influence appears to flow through the structural portfolio priority similarity captured by the RBO metric, reflecting shared issue prioritization arising from the broader lobbyist network built by revolving door dynamics [21, 14]. Scott [19] provides supporting evidence that peer influence in agenda development operates through shared information and trust among lobbyists rather than direct coordination. This mechanism supports our analysis that influence is recoverable from agenda alignment, even without traceable intermediary connections. We return to the implications of this finding in the Discussion.

4 Discussion

4.1 Case Studies: AbbVie and Oracle

The Oracle and AbbVie case studies together trace the full range of influence positions available in the directed network and ground the paper’s quantitative findings in traceable cascading firm behavior. AbbVie is the top within-community

agenda-setter in Healthcare/Pharmaceuticals for the 116th Congress, ranking 4th globally with *net_strength* = 4.167. Its bill portfolio is concentrated: 34 bills organized around drug pricing reform, pharmaceutical regulation, and COVID-era healthcare legislation. H.R.3 (the Lower Drug Costs Now Act), S.1895 (the Lower Health Care Costs Act), and S.2543 (the Prescription Drug Pricing Reduction Act) form the core of its agenda, and AbbVie files first on all three before any other Fortune 500 firm in its community. Under Hall and Deardorff’s (2006) legislative subsidy framework, this early positioning confers informational advantage: AbbVie invested deeply in policy-relevant expertise on drug pricing, and that investment made it a natural first-mover on the bills most central to that domain during the 116th Congress in particular

The cascading effects are visible throughout the Healthcare/Pharmaceuticals community. AbbVie’s strongest follower, Merck, shares 9 bills and an RBO of 0.257 with AbbVie and follows by a net temporal margin of +7 first-mover credits. AbbVie also leads Biogen (+4 net temporal), Humana, Johnson and Johnson, and Mutual of Omaha (+3 each), and Amgen, Gilead Sciences, Regeneron, and Walgreens (+2 each). The bill adoption cascade analysis confirms the timing: Biogen begins lobbying H.R.19 and S.2543 within one quarter of AbbVie’s first appearance on both bills. Gilead Sciences adopts H.R.2740, S.1416, and S.1895 within one quarter of AbbVie. This sequence illustrates Scott’s (2013) dyadic peer-influence mechanism: organizations with more prior legislative choices in common are more likely to adopt each other’s subsequent agenda selections, a pattern distinct from the bandwagon response in which all firms converge on bills that have gained broad legislative momentum regardless of their specific portfolio alignment.

net_strength does not, however, reflect a static firm attribute. AbbVie was a net follower in the 113th and 114th Congresses, rose through the 115th, and peaked in the 116th. The ascent coincides with drug pricing reform becoming the defining legislative priority of the 116th Congress, accelerated by the COVID-19 pandemic and the resulting national focus on pharmaceutical access. When a firm’s core regulatory domain becomes the session’s dominant legislative concern, that firm gains agenda-setting influence precisely because its existing industry knowledge and portfolio align with the bills that all other firms in the community subsequently need to lobby. The AbbVie trajectory is thus a micro-level illustration of the broader top-30 rotation finding: agenda-setting status is partly issue-driven, and the firms best positioned around a session’s most contested legislative issues emerge as the period’s top influencers.

Oracle presents the contrasting case. It ranks 290th of 293 firms globally on agenda-setting in the 116th Congress, with *net_strength* = −5.23, and ranks last within Tech/Telecom. This is not a product of disengagement with the legislative session: Oracle has 78 unique network neighbors and lobbies for 29 bills spanning high-skilled immigration reform, trade policy, appropriations, and COVID relief. Its portfolio is comparable to Microsoft’s, and the two share 8 bills with sufficient overlap to generate a directed edge weight of 1.786, which is one of the strongest within-community pairs. What Oracle lacks is first-mover advantage. Microsoft

filed first on 6 out of their 8 shared bills. Oracle’s negative *net_strength* reflects a systematic timing disadvantage instead of sparse connectivity, being consistently the second firm to arrive on bills already prioritized by Microsoft, AT&T, and other Tech/Telecom agenda-setting leaders.

The case of Oracle directly illustrates why centrality-based measures of influence would mischaracterize this firm. Its degree and weighted centrality in the bill affiliation network reflect its broad bill coverage; its *net_strength* captures the direction of its relationships with other firms, and that direction reflects it is a net-follower. The two measures recover fundamentally different properties. Oracle’s trajectory across congressional sessions further reinforces this distinction: in the 113th Congress, Oracle reached global rank 1 by *net_strength*, when its concentrated lobbying on tech-sector legislation moved ahead of its peers. By the 116th and 117th Congresses, the composition of Oracle’s agenda had diversified and its timing had slipped, dropping it to the bottom of the Tech/Telecom hierarchy. The same firm, in different sessions, occupies both extremes of the influence distribution. The measure is not capturing a fixed firm attribute but rather a session-specific position that responds to strategy, issue salience, and relative timing.

4.2 Varied Evidence of Strategic Complementarity

The strategic complementarity result for high-RBO pairs provides the first observational application of the Ballester et al. [1] strategic complementarity framework to corporate lobbying data. Their study predicts that in a network game with strategic complements, equilibrium effort levels are proportional to Bonacich centrality computed on the network of local complementarities. The RBO network is the natural realization of their complementarity graph. Edge weights encode portfolio alignment, the observable proxy for payoff spillovers between two firms’ lobbying activities. Under their framework’s equilibrium with stable complementarity structure, the influence hierarchy is self-reinforcing: followers’ best responses amplify the agenda-setter’s position, which in turn increases the setter’s *net_strength* in the next period. Our agenda-setting edge direction consistency reflects this through observable data, especially in sectors with legislated regulations.

The conditional structure of the result is important to understanding these findings. Strategic complementarity is not a universal property of all lobbying pairs. It holds only for firms in the top quartile of ranked portfolio similarity. Ward [22] identifies strategic offsetting (strategic substitution) as an equilibrium feature of a formal lobbying game in which competing groups exert opposing pressure on a policy outcome even without an explicit complementarity condition. Our results clarify where each operates. The AbbVie-Merck relationship is a high-RBO pair operating under strategic complementarity, each amplifying the other’s drug pricing agenda. A Healthcare/Pharmaceuticals firm lobbying against a Tech/Telecom firm on an appropriations bill, on the other hand, would be a low-RBO pair operating under substitutive complementarity.

One caveat remains under this interpretation. The lagged-RBO robustness test, which uses 115th Congress portfolio similarity to predict 116th Congress spending responses, does not reproduce the positive complementarity result. This raises the concern that the positive regressor for high-RBO firm pairs is partly driven by endogeneity, where high-RBO pairs are defined on the same data as the outcome variable. A future design using predetermined or externally determined portfolio similarity measures would be necessary to establish the causal interpretation of the complementarity result with greater confidence.

4.3 What Centrality Cannot Capture

The standard approach in corporate network research uses structural centrality as the proxy for inter-firm influence. Mizruchi and Bunting [18] provide the foundational validation of this approach, showing that centrality in corporate interlock networks predicts firm influence as assessed against historical accounts of corporate power – evidence that network position independently shapes influence outcomes beyond raw firm resources. However, Chu and Davis [8] document that the board interlock network has structurally collapsed since the late 1990s, motivating a search for alternative network instruments to recover similar information. The co-lobbying bill affiliation network is one candidate. However, our analysis of centrality measures show that this network is also insufficient, and it misses the temporal first-mover advantage and subsequent cascading effects revealed by the directed RBO network.

The distinction between structural position and temporal precedence in setting lobbying agendas corresponds to an important difference in what drives lobbying influence. Occupying a shared legislative space is captured by centrality, but *net_strength* is able to capture nodes corporations that occupy a leaders position within that space. Lorenz [17] shows that the composition of lobbying coalitions predicts committee consideration of bills, linking coalition structure to legislative outcomes. Our results extend this insight, suggesting that within any coalition, the firm that arrives first holds an asymmetric position that shapes the subsequent bill adoption of its coalition partners. That asymmetry is not visible in a static affiliation network.

4.4 Sector Heterogeneity and the Persistence of Influence

The divergence in Kendall’s *W* across communities shows that the stability of lobbying influence hierarchies is sector-specific. This finding advances the analysis of persistent versus transient lobbying actors from the lobbyist level, where LaPira et al. [15] established the two-Washingtons framework, to the client-firm level. Their distinction between institutionally embedded actors with cross-issue legislative agendas and transient actors engaged in short issue-specific campaigns maps directly onto our sectoral pattern. We find that Energy and Defense firms drive consistent legislative agendas on infrastructure, energy procurement, and environmental regulation across sessions, while Healthcare/Pharmaceuticals and

Finance/Insurance firms are more reactive to the specific priorities of each congressional cycle.

Heaney’s [11] analysis of health-policy lobbying provides a direct parallel for the Healthcare/Pharmaceuticals finding. His characterization of that sector as organized around broker actors (such as trade associations) who mediate inter-coalition dynamics aligns with our finding that no Healthcare/Pharmaceuticals firm reaches the persistent-leader threshold across our panel. The AbbVie peak in the 116th Congress illustrates the mechanism. The unusually concentrated legislative agenda around drug pricing amplified by COVID temporarily created the agenda conditions under which a single firm could dominate. The return of Merck to the top of the Healthcare/Pharmaceuticals ranking in the 117th Congress, as that concentrated drug pricing agenda receded, is consistent with the broker-mediated volatility that characterizes the sector across most sessions.

4.5 Mechanisms Beyond Shared Lobbyists

The failure of the shared-lobbyist mechanism reframes a long-standing conjecture in the lobbying literature. Prior work has speculated that shared human lobbyists function as information bridges between clients, with a firm’s bill priorities diffusing to peers through common hired contractors (Heaney, 2006; Carpenter, Esterling, and Lazer, 2004). Our results indicate that this direct relay mechanism cannot account for the agenda-setting patterns we observe. Among the 3,347 decisive influencer-follower pairs, shared lobbyist affiliation on the specific bill is essentially absent.

The failure of the direct communication hypothesis points toward a more fundamental structural channel: the latent social network among lobbyists themselves. LaPira and Thomas [14] characterize the revolving door in lobbying as a structural feature of the Washington lobbying ecosystem that creates this dense latent social network among lobbyists. Carpenter et al. [4] demonstrate that policy information in Washington lobbying networks flows primarily through weak-tie acquaintance relationships rather than directly hired contractors. It is weak ties between lobbyists, grounded in trust and social capital, that carry legislative priorities across organizational boundaries rather than direct shared-employment contracts with a common client. Scott [19] extends this notion, showing that agenda overlap develops through overlapping legislative choices shaped by shared informational environments rather than direct coordination. Together, these accounts suggest that the operative transmission channel behind the RBO influence signal is the broader inter-lobbyist trust network, which is invisible from LDA filings alone but theoretically central. Directly measuring the weak-tie structure of the lobbyist social network and linking it to firm-level agenda-setting patterns identified here represents a tractable and high-priority direction for future empirical work.

Trade associations are a further indirect channel. Brulle and Downie [3] document that trade association lobbying on climate issues represents a substantial and under-measured share of total business political activity, which is evidence

that LDA-based analyses systematically undercount coordinated corporate political spending channeled through associations whose filings are separate from client-level reports. Under this channel, shared trade association membership transmits policy priorities across firms without generating observable shared-lobbyist records at the bill level. Kowal [13] extends this further, showing that corporate political behavior — including lobbying activity and campaign contributions — diffuses through trade association networks: firms adopt lobbying postures that mirror those of their trade-association-connected peers, a form of coordinated imitation that leaves no direct trace in client-level LDA filings. This peer-imitation channel is therefore invisible to any analysis restricted to direct lobbying reports alone. Together, these findings suggest that the RBO signal we recover from client-level data captures only a portion of the coordinated political activity among Fortune 500 firms; trade-association-mediated coordination represents a complementary and empirically unmeasured layer of strategic interaction.

A related structural finding concerns the cross-sector dimension of the influence network. Of the 1,890 directed edges in the 116th Congress influence network, 44.1% cross Leiden community boundaries. This pattern extends and refines Carpenter, Esterling, and Lazer’s (1998) weak-ties argument. The findings show that bridging ties across industry communities exist and carry consistent directional signal, but they are concentrated in appropriations and budget space, where Defense and Industrial firms take the lead. Cross-community influence is real, but it is issue-specific rather than widely spread.

The practical implication is that the causal mechanism behind the RBO influence signal cannot be identified from LDA data alone. The falsification of the direct relay hypothesis redirects attention to indirect lobbyist-network information diffusion and trade association intermediation as the probable channels. Testing these channels requires data beyond what LDA filings provide.

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